the two planets. A result of importance to everyone on Earth emerged from what might well have seemed the most blue-sky, abstract, impractical kind of work, understanding the chemistry of minor constituents in the upper atmosphere of another world.

There's also a Mars connection. With *Viking* we found the surface of Mars to be apparently lifeless and remarkably deficient even in simple organic molecules. But simple organic molecules *ought* to be there, because of the impact of organic-rich meteorites from the nearby asteroid belt. This deficiency is widely attributed to the lack of ozone on Mars. The *Viking* microbiology experiments found that organic matter carried from Earth to Mars and sprinkled on Martian surface dust is quickly oxidized and destroyed. The materials in the dust that do the destruction are molecules something like hydrogen peroxide—which we use as an antiseptic because it kills microbes by oxidizing them. Ultraviolet light from the Sun strikes the surface of Mars unimpeded by an ozone layer; if any organic matter *were* there, it would be quickly destroyed by the ultraviolet light itself and its oxidation products. Thus part of the reason the topmost layers of Martian soil are antiseptic is that Mars has an ozone hole of planetary dimensions—by itself a useful cautionary tale for us, who are busily thinning and puncturing *our* ozone layer.

(2) Global warming is predicted to follow from the increasing greenhouse effect caused largely by carbon dioxide generated in the burning of fossil fuels—but also from the buildup of other infrared-absorbing gases (oxides of nitrogen, methane, those same CFCs, and other molecules).

Suppose that we have a three-dimensional general circulation computer model of the Earth's climate. Its programmers claim it's able to predict what the Earth will be like if there's more of one atmospheric constituent or less of another. The model does very well at "predicting" the present climate. But there is a nagging worry: The model has been "tuned" so it will come out right—that is, certain adjustable parameters are chosen, not from first principles of physics, but to get the right answer. This is not exactly cheating, but if we apply the same computer model to rather different climatic regimes—deep global warming, for instance—the tuning